

Inputs

model matrix
 X ($n \times p$),
residuals, weights

$A = X'X = R'R$
`crossprod(x); chol()`
one factorization, $p \times p$

$\omega = (\hat{e}_t^2 g_t)$, length n
`x * omega`: $n \times p$ row-scaled
 $M = X' \text{diag}(\omega) X$
`crossprod(x, x * omega)`, $p \times p$

Solve, no inverse stored

solve $AL = M$,
then $AZ = L'$
 A^{-1} applied twice via
four triangular solves:
2 forward, 2 backward,
 p -column right sides
 $\hat{\Psi}_{\text{HC}} = A^{-1} M A^{-1}$
symmetrized, $p \times p$

Objects deliberately not formed: dense $\text{diag}(\omega)$ ($n \times n$), explicit X' , explicit A^{-1} , and $A^{-1}X'$ ($p \times n$).